

LA TUQUE, QC, Canada

WMO: 713780

Lat: 47.411N

Lon: 72.787W

Elev: 554

StdP: 14.40

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-18.8	-30.7	1.0	-23.3	-25.6	1.3	-17.9	18.0	13.6	16.2	13.4	1.7	110	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	86.3	66.8	82.7	65.0	79.6	63.8	71.0	79.3	69.1	76.9	67.3	74.7	8.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	106.5	74.5	66.5	99.8	72.4	64.7	93.4	70.4	35.3	79.1	33.6	76.7	32.1	74.5	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.4	12.8	DB	-32.6	92.7	4.9	3.2	-36.1	95.0	-39.0	96.9	-41.7	98.8	-45.2	101.1	(4)
(5)			WB	-32.9	74.6	4.9	1.9	-36.4	76.0	-39.3	77.1	-42.1	78.2	-45.7	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.7	6.6	9.9	21.9	36.6	51.5	61.3	66.0	63.8	55.8	43.0	29.9	15.7		
	DBStd	22.81	14.17	12.95	12.38	8.42	8.06	7.24	5.47	5.87	7.71	7.90	9.72	12.68		
	HDD50	5768	1346	1122	873	406	78	4	0	0	27	244	604	1064		
	HDD65	9824	1811	1542	1337	851	424	156	54	94	292	681	1053	1529		
	CDD50	1626	0	0	0	5	125	342	496	427	201	29	1	0		
	CDD65	206	0	0	0	0	6	44	84	56	16	0	0	0		
	CDH74	2364	0	0	1	4	184	596	874	546	153	6	0	0		
	CDH80	641	0	0	0	0	46	190	241	125	38	1	0	0		
	WSAvg	5.2	5.1	5.2	6.0	6.1	5.6	5.0	4.7	4.4	4.6	5.0	5.3	4.9		
	PrecAvg	37.8	2.2	2.3	2.4	2.4	3.0	3.6	4.7	4.1	3.8	3.0	3.3	3.4		
	PrecMax	48.7	3.8	5.0	5.9	4.6	9.9	6.6	11.6	6.0	6.6	5.9	7.8	5.7		
(15) Precipitation	PrecMin	33.2	0.8	0.1	0.5	0.7	0.6	2.0	2.1	1.4	1.9	1.3	1.4	1.2		
	PrecStd	4.2	0.7	1.5	1.1	0.9	1.9	1.2	2.1	1.3	1.2	1.2	1.3	1.2		
	DB	43.6	42.1	54.7	69.9	85.7	90.3	90.6	88.1	84.7	71.9	57.7	43.1	(19)		
	MCWB	42.4	37.5	43.4	52.9	60.6	67.6	70.0	67.4	67.6	58.8	53.5	40.9	(20)		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	36.9	36.8	47.0	62.5	79.1	85.4	85.8	83.4	78.1	65.1	51.6	38.3	(21)		
	2% MCWB	35.5	33.2	38.5	47.6	58.0	65.7	67.9	66.1	65.3	55.9	47.4	36.5	(22)		
	5% DB	33.0	33.7	42.4	56.8	73.9	80.9	82.2	80.0	73.6	59.8	47.3	35.1	(23)		
	5% MCWB	31.4	31.2	35.8	44.6	55.8	63.1	65.6	65.0	62.6	53.8	43.8	33.5	(24)		
	10% DB	28.1	30.8	38.5	51.5	68.6	76.4	79.0	76.5	69.5	55.5	43.0	32.8	(25)		
	10% MCWB	26.4	28.4	33.8	42.1	53.9	60.9	64.3	63.6	61.2	50.2	39.9	31.3	(26)		
	0.4% WB	42.5	39.4	46.5	56.6	66.8	72.7	74.2	71.9	71.1	62.1	54.8	41.3	(27)		
	0.4% MCDB	43.6	41.2	50.7	65.8	75.5	83.7	83.8	80.3	79.7	68.3	57.8	42.5	(28)		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	35.7	34.2	39.6	50.0	62.8	69.4	71.5	69.8	67.9	58.1	48.6	36.8	(29)		
	2% MCDB	36.7	35.9	44.2	56.8	72.4	78.2	79.3	77.7	74.6	63.4	50.8	38.1	(30)		
	5% WB	31.5	31.6	36.8	46.4	59.6	66.7	69.3	68.0	65.1	54.7	44.1	33.9	(31)		
	5% MCDB	33.0	33.7	41.3	53.4	67.7	75.0	77.2	75.5	71.4	59.3	46.8	34.9	(32)		
	10% WB	26.5	28.3	34.3	43.1	56.3	64.4	67.3	66.1	62.4	51.4	40.5	31.5	(33)		
	10% MCDB	28.0	30.9	37.9	50.2	65.7	72.5	74.8	72.4	68.2	54.8	42.9	32.7	(34)		
(35) Mean Daily Temperature Range	MDBR	20.9	23.7	22.4	21.0	25.1	23.9	22.1	21.9	20.9	16.8	13.9	16.0	(35)		
	5% DB	18.3	18.1	24.7	33.1	35.4	31.5	27.8	26.9	26.0	24.3	19.5	13.1	(36)		
	5% WB	17.9	15.6	16.8	19.1	16.6	13.7	12.2	12.0	14.0	15.4	16.1	12.4	(37)		
	MCDBR	18.1	16.8	20.8	26.6	25.9	23.0	22.7	21.1	22.4	20.8	17.9	12.5	(38)		
	MCWBR	17.8	15.0	15.2	17.5	14.8	12.3	11.9	11.1	13.8	14.6	16.0	12.4	(39)		
(40) Clear-Sky Solar Irradiance	taub	0.270	0.278	0.309	0.341	0.361	0.388	0.396	0.386	0.365	0.336	0.302	0.263	(40)		
	taud	2.268	2.245	2.230	2.378	2.422	2.351	2.334	2.371	2.423	2.495	2.486	2.360	(41)		
	Ebn at Noon	257	282	288	290	286	278	273	272	267	259	247	248	(42)		
	Edh at Noon	24	31	37	36	35	38	38	36	31	24	20	20	(43)		
(44) All-Sky Solar Radiation	RadAvg	428	742	1161	1463	1618	1741	1674	1457	1125	651	370	292	(44)		
	RadStd	69	112	112	124	160	136	169	76	90	59	49	23	(45)		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page